

Design Project Presentation

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Description & Introduction to the stakeholders' problem

A corkboard has multiple different names, for example, some people call it a bulletin board, a pin board, or a notice board. This board is a physical surface that is used to display messages, to show schedules, to celebrate achievements, to be a decoration, to display photos, or for inspiration. Typically mounted on a wall, it is made of materials like cork, foam, or fabric that allow you to easily attach and rearrange items using push pins or thumbtacks. The board is commonly used in a classroom for teachers and students. Also, they serve as a communication tool, a learning tool, and a way to make the classroom environment more engaging and organized. In most teachers' classrooms, papers are sometimes taped to walls or placed on desks, which can cause them to fall or get lost and make the classroom disorganized. This is a decent-sized issue and problem for most teachers. This problem happens because some schools don't provide pin boards in classrooms when they are first built, or the pinboard may sometimes get messed up or become old. Due to this being a problem for a teacher in our school, we decided to construct a pinboard for him. He stated to us that it was a problem for him because he wanted to be extremely organized. He would be our main stakeholder for the creation of the pinboard, but students would also be considered stakeholders because they rely on the posted information to stay on track. The School and other staff benefit as well because having a pinboard can reduce wall damage.

Prior Design Attempts & Design Requirements

Many different companies have created different styles of pin boards before. Due to the pinboard not being our personal invention, we decided to look up prior design attempts. Two brands we took ideas from were Foray and Lockways. Both boards were rectangular, made of cork, had an aluminum frame around the perimeter, and featured the ability to self-heal when push pins or thumbtacks are pushed into the board. The prior design attempts helped us gain information on how we wanted to construct our pinboard. Also, this analysis helped us out because it gave us ideas on where to buy materials from and it showed us how the finished product should look. After getting ideas on where to start, a pivotal thing we had to do was come up with design requirements in order to work towards a solution. Every design has a certain number of requirements in order to start designing it. In our design, 7 things are considered requirements, whether they are high-priority or low-priority. The five high-priority requirements are adequate cork thickness and density, high-quality self-healing material, stable mounting, durable frame, and size and shape. The two low-priority requirements are environmental resistance and aesthetic versatility. Although we ranked the requirements the way we did, all of them matter towards the creation of the project to come up with a solution for our stakeholders' problem.

Manufacturing Methods & STEM

Coming up with manufacturing methods or steps to complete the project was an important part of our creation process. First, the most important step is getting the materials together. Our total price was around \$110 dollars. The second step is to measure and mark the corkboard to fit the desired place of the stakeholder. Third, mark where we wanted to cut the aluminum frame pieces. Fourth, make cuts on the corkboard and the aluminum frame pieces. Then, place adhesive strips along the aluminum brackets. After, we need to place the frame around the perimeter of the corkboard. Lastly, place the adhesive strips along the back of the corkboard to finally hang it up in the desired place for the stakeholder. These 7 steps are important because it would be a failed project if we didn't follow them. STEM plays a major part in our project because we used science, technology, engineering, and math to create the board. For example, the science portion focuses on the material properties, the technology is with the tools and manufacturing, the engineering portion is the whole design process, and mathematics focuses on the measurements, prices, etc. Ultimately, we used the STEM concepts to justify our design decisions. It's important that the STEM Principles and Practices that are discussed between our group are actually appropriate and useful to the problem.

Data Testing Plan & Results

Each project that is created by engineers usually has at least one test sample, which allows you to collect objective, measurable data. Also, each project has testable and non-testable attributes. An example of a testable attribute is the self-healing surface, while a non-testable attribute is its long-term environmental resistance. We had to get the data from the actual finished product because we only created one sample. First, we had to come up with a data testing plan. Each high-priority requirement should be tested during the engineering process. We tested its thickness, its self-healing ability, how stable the mounting was, and how durable the frame was. For each one, we had to come up with a testing procedure, test cases and a number of trials, expected range of results, ultimately determining whether the test would pass or fail. For thickness and self-healing ability we punctured holes to determine if they would close. For the stable mounting test, we pulled and shook the board to see if there was any shifting, tilting, or loosening. Lastly, we hit and tried to be the frame to see if it would come out of place. After coming up with a data plan, we tested the plan. Every high-priority requirement went smoothly except the stable mounting. The only reason this somewhat failed because we mounted it with adhesive strips instead of screwing the board into the wall. The data helped us learn what to do again if we reconstruct and what not to do when reconstructing. Overall, the data testing plan went well and we learned from our mistakes.

Stakeholders' Feedback

Another major point of this project was getting feedback from the stakeholder. In summary, Mr. Benda stated that we did a great job constructing the board for it to be our first engineering project. He liked how the cork is pretty thick and self-healing, which allows him to hang up important things that he needs to hang. He said that, in the future, we should hang it up using screws and/or braces, so that it can be stronger and able to last longer than the adhesive strips. Another alternative he gave to us is to make a test model using cheaper materials to use it as a test run or a sample project to help us figure out what is needed to make it near perfect. Overall, he liked the test results.



Conclusion

In conclusion, throughout this project, we learned that the engineering design process requires research, planning, teamwork, testing, and improvement. Defining our stakeholders' problem early helped guide our design, while researching existing cork boards gave us ideas for materials and construction. We also learned that design requirements are important because they help engineers focus on the most necessary features, such as durability and stable mounting. Testing showed us that even good ideas can fail, since our mounting method using adhesive strips was not as stable as expected. This taught us that engineers must learn from mistakes and improve their designs based on test results and stakeholder feedback. Overall, the project showed us that engineering is a step-by-step process that involves problem-solving, communication, and continuous improvement. If we continued this project, we would improve the mounting system by using screws or stronger wall anchors instead of adhesive strips because our testing showed the board was not stable enough. We would also create a smaller test sample before building the final corkboard so we could identify problems earlier and improve the final product. If we did this project again, we would keep our research process and high-priority design requirements because they helped us create a durable and organized cork board that satisfied the stakeholder. We would also improve teamwork by dividing responsibilities more evenly and communicating more throughout the project. Having more time, additional materials, and more testing opportunities would have made the project easier to complete and would have allowed us to make an even stronger final design.

Full Write-Up Put Together

https://docs.google.com/document/d/13TS0sqaUkTnphVTqSs-YHEATBAHOOwx_Gtu7tZ06Vlo/edit?usp=sharing